MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY,
STYLES CWR06 AND CWR09

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-55365.

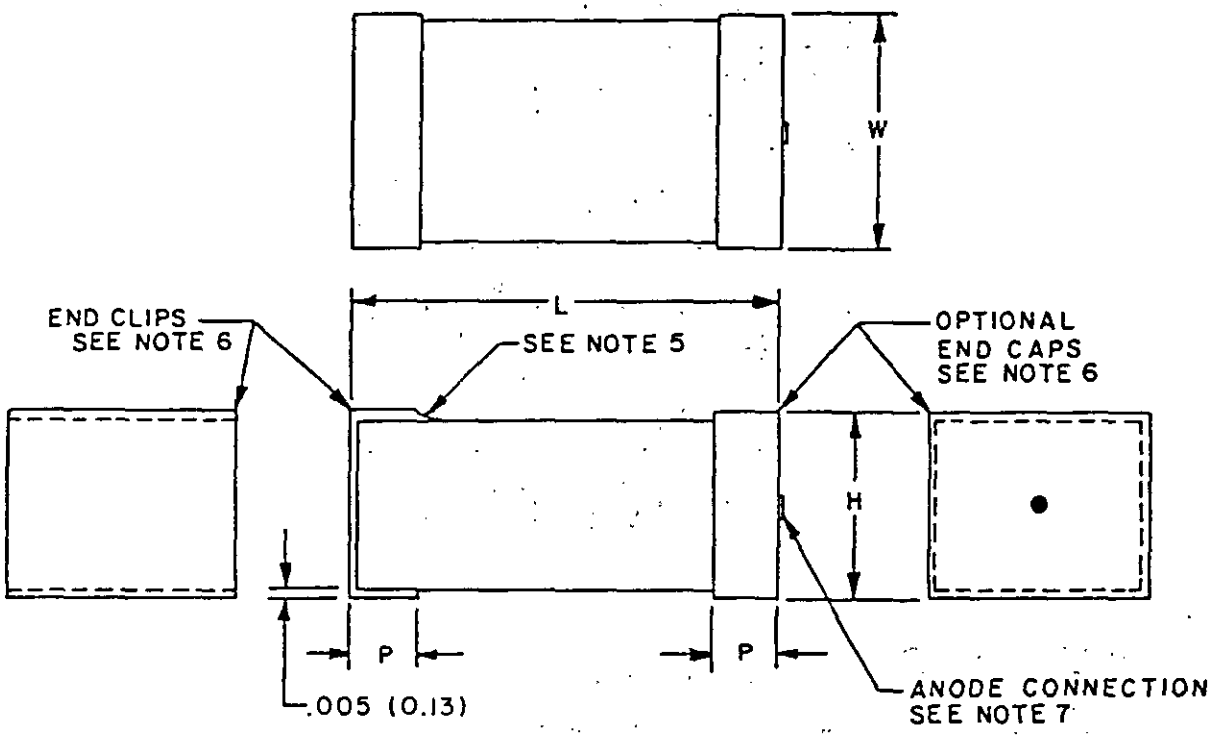


FIGURE 1. Style CWR06 - (encapsulated) capacitors.

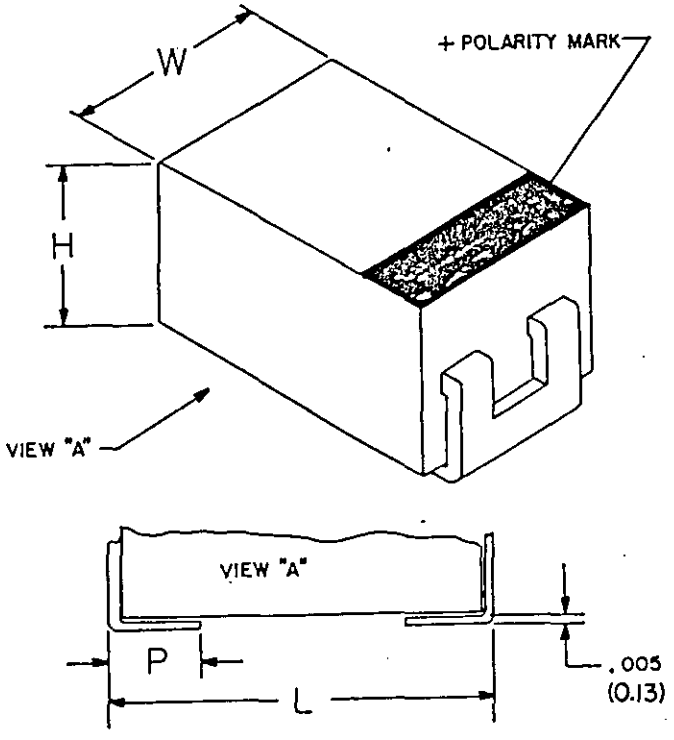
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Case size	L	W	H	P
	±.015 (0.38)	±.015 (0.38)	±.015 (0.38)	±.005 (0.13)
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3 Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
5. Cathode terminal may be epoxy coated on one surface for polarity identification.
6. Anode and cathode terminals may be two-sided, three-sided, or five-sided, and may not be the same on some designs, but will meet all specification requirements.
7. The anode terminal shall be identified by the riser wire connection, which may extend the case size .015 (0.38 mm) maximum.
8. When solder coated terminations are required add .015 (0.38 mm) to dimensions.

FIGURE 1. Style CWR06 (encapsulated) capacitors - Continued.

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Case size	L ±.015 (.38)	W ±.015 (.38)	H ±.015 (.38)	P ±.005 (.13)
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (.76)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
5. Cathode terminal may be epoxy coated on one surface for polarity identification.
6. Anode and cathode terminals may be two-sided, three-sided, or five-sided, any may not be the same on some designs, but will meet all specification requirements.
7. The anode terminal shall be identified by the riser wire connection, which may extend the case size .015 (0.38 mm) maximum.
8. When solder coated terminations are required add .015 (0.38 mm) to dimensions.

FIGURE 2. Style CWR09 molded chip.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Chip.

Material: Sintered tantalum.

Leads: Unleaded.

Termination finish: In accordance with MIL-C-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-C-55365).

Operating temperature range: -55°C to +125°C.

Failure rate level: B (0.1 percent), C (0.01 percent), D (0.001 percent), E (0.0001 percent), M (1.0 percent), P (0.1 percent), R (0.01 percent), and S (0.001 percent).

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition E.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):	DCL	-	See table I.
	Cap.	-	Shall not exceed tolerance of table I value.
	DF	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Shall not exceed ±10 percent of initial measurement.
	DF	-	See table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±10 percent of initial +25°C measurement.
	DF	-	See table I.
Step 5 (+125°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±15 percent of initial +25°C measurement.
	DF	-	See table I.
Step 6 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.

Surge voltage: In accordance with MIL-C-55365.

DCL	-	See table I.
ΔCap.	-	Within ±5 percent of initial value.
DF	-	See table I.

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Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C: DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial measured value.
DF - See table I.

At 85°C: DCL - See table I.

At 125°C: DCL - See table I.

10,000 hours:

At 25°C: DCL - 200 percent of initial requirement.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
DF - See table I.

At 85°C: DCL - 200 percent of initial requirement.

At 125°C: DCL - 200 percent of initial requirement.

TABLE I. Style CWR06 characteristics.

Type designation 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	µF	µA	µA	µA	%	%	%	
CWR06C-225--	4	2.2	1.0	10	12	6	8	8	A
CWR06C-475--	4	4.7	1.0	10	12	6	8	8	B
CWR06C-635--	4	6.8	1.0	10	12	6	8	8	C
CWR06C-106--	4	10.0	1.0	10	12	8	8	10	D
CWR06C-156--	4	15.0	1.0	10	12	8	10	12	E
CWR06C-336--	4	33.0	2.0	20	24	8	10	12	F
CWR06C-686--	4	68.0	3.0	30	36	10	12	12	G
CWR06C-107--	4	100.0	4.0	40	48	10	12	12	H
CWR06D-155--	6	1.5	1.0	10	12	6	8	8	A
CWR06D-335--	6	3.3	1.0	10	12	6	8	8	B
CWR06D-475--	6	4.7	1.0	10	12	6	8	8	C
CWR06D-685--	6	6.8	1.0	10	12	6	8	8	D
CWR06D-106--	6	10.0	1.0	10	12	8	10	12	E
CWR06D-226--	6	22.0	2.0	20	24	8	10	12	F
CWR06D-476--	6	47.0	3.0	30	36	10	12	12	G
CWR06D-686--	6	68.0	4.0	40	48	10	12	12	H
CWR06F-105--	10	1.0	1.0	10	12	6	8	8	A
CWR06F-225--	10	2.2	1.0	10	12	6	8	8	B
CWR06F-335--	10	3.3	1.0	10	12	6	8	8	C
CWR06F-475--	10	4.7	1.0	10	12	6	8	8	D
CWR06F-685--	10	6.8	1.0	10	12	6	8	8	E
CWR06F-156--	10	15.0	2.0	20	24	8	8	10	F
CWR06F-336--	10	33.0	3.0	30	36	10	12	12	G
CWR06F-476--	10	47.0	5.0	50	60	10	12	12	H
CWR06H-684--	15	.68	1.0	10	12	6	8	8	A
CWR06H-155--	15	1.5	1.0	10	12	6	8	8	B

See footnote at end of table.

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TABLE I. Style CWR06 characteristics - Continued.

Type designation 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	µF	µA	µA	µA	%	%	%	
CWR06H-225--	15	2.2	1.0	10	12	6	8	8	C
CWR06H-335--	15	3.3	1.0	10	12	6	8	8	D
CWR06H-475--	15	4.7	1.0	10	12	6	8	8	E
CWR06H-106--	15	10.0	2.0	20	24	6	8	8	F
CWR06H-226--	15	22.0	4.0	40	48	8	8	10	G
CWR06H-336--	15	33.0	5.0	50	60	8	8	10	H
CWR06J-474--	20	.47	1.0	10	12	6	8	8	A
CWR06J-684--	20	.68	1.0	10	12	6	8	8	B
CWR06J-105--	20	1.0	1.0	10	12	6	8	8	B
CWR06J-155--	20	1.5	1.0	10	12	6	8	8	C
CWR06J-225--	20	2.2	1.0	10	12	6	8	8	D
CWR06J-335--	20	3.3	1.0	10	12	6	8	8	E
CWR06J-685--	20	6.8	2.0	20	24	6	8	8	F
CWR06J-156--	20	15.0	3.0	30	36	6	8	8	G
CWR06J-226--	20	22.0	4.0	40	48	6	8	8	H
CWR06K-334--	25	.33	1.0	10	12	6	8	8	A
CWR06K-684--	25	.68	1.0	10	12	6	8	8	B
CWR06K-105--	25	1.0	1.0	10	12	6	8	8	C
CWR06K-155--	25	1.5	1.0	10	12	6	8	8	D
CWR06K-225--	25	2.2	1.0	10	12	6	8	8	E
CWR06K-475--	25	4.7	2.0	20	24	6	8	8	F
CWR06K-685--	25	6.8	2.0	20	24	6	8	8	G
CWR06K-106--	25	10.0	3.0	30	36	6	8	8	G
CWR06K-156--	25	15.0	4.0	40	48	6	8	8	H
CWR06M-224--	35	.22	1.0	10	12	6	8	8	A
CWR06M-474--	35	.47	1.0	10	12	6	8	8	B
CWR06M-684--	35	.68	1.0	10	12	6	8	8	C
CWR06M-105--	35	1.0	1.0	10	12	6	8	8	D
CWR06M-155--	35	1.5	1.0	10	12	6	8	8	E
CWR06M-335--	35	3.3	1.0	10	12	6	8	8	F
CWR06M-475--	35	4.7	2.0	20	24	6	8	8	G
CWR06M-685--	35	6.8	3.0	30	36	6	8	8	H
CWR06N-104--	50	.10	1.0	10	12	6	8	8	A
CWR06N-154--	50	.15	1.0	10	12	6	8	8	A
CWR06N-224--	50	.22	1.0	10	12	6	8	8	B
CWR06N-334--	50	.33	1.0	10	12	6	8	8	B
CWR06N-474--	50	.47	1.0	10	12	6	8	8	C
CWR06N-684--	50	.68	1.0	10	12	6	8	8	D
CWR06N-105--	50	1.0	1.0	10	12	6	8	8	E
CWR06N-155--	50	1.5	1.0	10	12	6	8	8	F
CWR06N-225--	50	2.2	2.0	20	24	6	8	8	F
CWR06N-335--	50	3.3	2.0	20	24	6	8	8	G
CWR06N-475--	50	4.7	3.0	30	36	6	8	8	H

1/ Complete type designation shall include additional symbols to indicate style, termination finish, capacitance tolerance, and failure rate level.

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Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 11, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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